ConceptBook: A Graph-First Framework for AI-Generated Curricula

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1 Abstract

We present ConceptBook, a framework in which teaching any subject reduces to two acts: *authoring* a concept-graph — a directed acyclic graph of primitive concepts, composition relationships, per-concept verifiers, and a target mastery concept — and *running* a single SPL workflow that converts that graph into a prerequisite-ordered, LLM-generated, verifier-checked textbook. No code changes are needed between domains. The curriculum specification is the graph; everything else — teaching order, content generation, content verification, and interactive navigation — is derived from it algorithmically. The graph can be authored in any form: a structured file, a GUI, or generated from domain knowledge — the framework is independent of the authoring tool.

We demonstrate the system across ten domains: Linear Algebra, Introductory Geometry, Classical Mechanics, Chemistry Elements, Chinese Characters, English Morphology, Python for Science, SageMath, Lean Theorem Proving, and Music Theory. Each domain is specified as a concept-graph (4–13 primitives, 14–25 concepts) and processed by an identical SPL workflow (`build_concept_book.spl`) that computes a topologically-ordered teaching sequence, generates each section via LLM, verifies it using a per-concept verifier (structural graph check, SymPy, Sage, or Lean), and assembles a self-contained HTML book with MathJax rendering. An interactive vis.js concept-graph navigator is generated in parallel.

Every domain in the system exhibits what we call the *LEGO payoff shape*: a small primitive brick set plus one composition principle unlocks the ability to decode or construct a space orders of magnitude larger — thousands of Chinese characters from ~200 radicals, most academic vocabulary from ~220 morphemes, every chemistry formula from ~20 elements and one bonding rule. We argue that this shape is not an artifact of our choice of domains but a structural property of well-organized knowledge, and that making it explicit in a machine-readable graph is what makes curriculum generation computable. The system, including all ten domain graphs, the SPL workflow, graph algorithms library, and web portal, is released as open source under Apache 2.0.

Keywords: educational AI, knowledge graphs, declarative curriculum, concept-book generation, structured prompt language, prerequisite ordering, LLM-generated content, verifier-checked pedagogy

2 Introduction

2.1 The Missing Layer in Educational AI

The current generation of AI tutoring systems divides into three families. Conversational tutors (Khanmigo, Socratic) engage learners in dialogue but carry no explicit model of what concepts depend on what. Adaptive

testing systems (Item Response Theory, Knewton) measure knowledge but do not generate it. Retrieval-augmented systems (RAG over textbooks) retrieve existing authored prose but cannot produce a coherent book from scratch because they have no model of the domain's internal structure.

The result is what we call *vibe tutoring*: AI that generates plausible-sounding explanations without any structural guarantee that prerequisites have been covered, that the teaching order is sound, or that the content is verifiable. What all three families lack is the same thing: an explicit, machine-readable prerequisite graph. Without it, teaching order is editorial judgment, content generation is unconstrained, and verification is absent. The gap is not a failure of the AI — it is a failure of representation.

2.2 The Central Claim

We claim that curriculum knowledge is fundamentally graph-structured, and that making the graph explicit is sufficient to make teaching computable. Specifically:

- **Teaching** = authoring a concept-graph
- **Teaching order** = topological sort of the graph, weighted by productivity
- **Content generation** = a parameterized SPL workflow over the sorted sequence
- **Content verification** = per-concept verifier dispatch (structural, SymPy, Sage, Lean)
- **Learning** = navigating the graph toward a target concept

The concept-graph is the complete curriculum specification. The SPL workflow `build_concept_book.spl` is domain-agnostic: it accepts any concept-graph conforming to the schema, and produces a verified, prerequisite-ordered textbook. Adding a domain means authoring one concept-graph. No code changes.

2.3 Origins: From Reductionism to Universal Curriculum

This framework did not begin as an educational system. It began as a physics-inspired question: what are the irreducible constituents of the Chinese writing system, and what composes from them?

A physicist's instinct is reductionist — find the smallest number of independent primitives from which everything else can be constructed. Applied to Chinese characters, this instinct asks: how many radicals (部首) are truly irreducible, and how do they compose into the ~50,000 characters of the writing system? The ZiNets project [CITE-ZiNets] answered this computationally: 422 elemental characters (元字), a directed composition graph, and a machine-checked reducibility theorem — the full writing system is reducible to this primitive set, and no strict subset suffices. The phono-semantic principle (形声字), which accounts for over 80% of all Chinese characters, emerged not as an editorial claim but as a theorem: the semantic pictograms (FORM) and the phonetic lenders (SOUND) are mutually irreducible, and their composition is the generative engine of the writing system.

Only later — after months of AI research building SPL [CITE-SPL] as a declarative language for LLM orchestration — did the deeper pattern become clear: Chinese characters are not a special case. They are the most transparent instance of a structure that is present in every well-organized domain. Physics has position, time, mass, and force — four primitives from which all of classical mechanics composes. Music has pitch, semitone, beat, and meter. English vocabulary has roots, prefixes, and suffixes. The Chinese writing system simply makes this structure *visible*, encoded into the calligraphic form of each character across three millennia of refinement.

The writing system is, in this sense, humanity's oldest working concept-book: a domain whose primitives and composition rules were so carefully curated that a learner who masters ~200 radicals and one principle can decode thousands of characters they have never seen. Every domain in this paper has the same structure. ConceptBook is the system that makes it runnable.

The mission that drives this work is simple: **make education accessible to everyone**. The expertise to teach a subject well — to know its primitives, its composition rules, its capstone — exists in the minds of domain experts worldwide. The barrier is not knowledge; it is the cost of turning that knowledge into a curriculum that a learner anywhere in the world can follow. A concept-graph and a single command should be enough.

2.4 The LEGO Payoff Shape

Every domain in our system exhibits the same payoff pattern:

> Learn a small brick set (the primitives) plus one composition principle (the capstone), and you can decode or construct a space orders of magnitude larger than what you explicitly memorized.

- Chinese characters: ~200 radicals + the phono-semantic principle → thousands of characters
- English morphology: ~220 morphemes (roots, prefixes, suffixes) + the decoding principle → most academic vocabulary
- Chemistry: ~20 elements + one bonding rule (valence electron) → every formula on every label
- Classical mechanics: 4 primitives (position, time, mass, force) + normal modes → orbital mechanics, robotics, quantum bridge
- Music theory: 4 primitives (pitch, semitone, beat, halving) + the four-chord loop → most popular music

We call this the *LEGO payoff shape* because it mirrors how LEGO works: a finite brick catalogue, a composition grammar, and a generative space far larger than the catalogue. The payoff shape is not a metaphor we impose — it is a theorem the `graph_lib.reducible()` function checks for each domain: the graph is reducible to its declared primitive set, and the first-radical subset alone is not sufficient.

2.5 Contributions

1. **Concept-graph schema** — a formal specification (primitives, composition edges, per-concept verifiers, capstone) as a complete curriculum description; authorable via structured file, GUI, or LLM-assisted interview — independent of representation format
2. **Domain-agnostic SPL workflow** — `build_concept_book.spl` runs against any conforming concept-graph, producing a verified, prerequisite-ordered book
3. **Graph-algorithm teaching order** — `productivity_order` and `learning_path` as reproducible, computable curriculum sequencing
4. **Cross-domain verifier dispatch** — structural, SymPy, Sage, and Lean verifiers unified behind a single `CALL verify_content(...)` in SPL
5. **Ten working domains** — spanning STEM mathematics, physics, computing tools, natural languages, and music, all running through the same pipeline
6. **Open-source portal** — an interactive web portal (vis.js navigator + FastAPI backend + GitHub Pages deployment) for all ten domains

3 Related Work

3.1 Concept Prerequisite Graphs: Mined vs. Authored

A substantial line of NLP/ML work attempts to *discover* prerequisite relationships automatically — from MOOC transcripts [ALSaad et al., EDM 2018], course enrollment patterns [Liu et al., JAIR 2016], Wikipedia links [Talukdar & Cohen, BEA 2012], and neural classifiers on Coursera data [Pan et al., ACL 2017]. A 2025 ACM Computing Surveys review covers the full landscape and identifies *cross-domain generalization* as the central open problem. The proposed system sidesteps this problem entirely by treating the prerequisite DAG as a human-authored input — domain experts specify the concept-graph; the algorithms assume it is correct and build from it. This shifts effort from mining to authoring, but removes all the noise and approximation that comes with automated extraction.

The most directly relevant precedent in this line is Loach & Wang (PLOS ONE, 2016), who model 6,990 Chinese characters as a structural DAG and apply a frequency-weighted topological sort to produce an optimal learning sequence — demonstrating empirically that topological ordering of a prerequisite DAG improves learning efficiency. Our `productivity_order` algorithm is a generalization of this idea to arbitrary domains.

3.2 Knowledge Space Theory and ALEKS

Doignon & Falmagne (1985) introduced Knowledge Space Theory (KST), which models learner knowledge as a subset of items drawn from a domain, with a family of "feasible" knowledge states closed under union. ALEKS implements KST at scale — adapting to millions of learners across mathematics and chemistry via $\leq$25 diagnostic questions covering exponentially many latent states. KST's "fringe" (the items one step beyond a learner's current frontier) is structurally analogous to our `graph_lib.gap(known, target)` function. The key contrast: KST is designed for *assessment* — modeling what a learner knows — while the proposed system is designed for *generation* — producing the content the learner reads. ALEKS does not generate explanations; it selects pre-authored problems. The proposed system does not track learner state; it generates prerequisite- ordered books. They address complementary halves of the same problem.

3.3 Intelligent Tutoring Systems

Anderson et al.'s Cognitive Tutor (1995) demonstrated that expert-authored domain models — encoded as production rules over knowledge components — achieve learning gains comparable to human tutoring. VanLehn's meta-analysis (2011) confirmed this at scale (d=0.76 for step-based ITS vs. d=0.79 for human tutors). The decisive limitation is authoring cost: a single Cognitive Tutor course requires months of expert effort per domain, making scaling prohibitive. ConceptBook is designed to make this authoring act as lightweight as possible — a domain expert who can describe their subject's primitives and composition rules can produce a concept-book in hours rather than months, using whatever authoring interface they prefer.

AutoTutor (Graesser et al., 2004) generates Socratic dialogue around hand-authored curriculum scripts; Piech et al.'s Deep Knowledge Tracing (NeurIPS 2015) replaces explicit knowledge components with LSTM-learned latent states. Neither generates expository prose; neither uses an explicit machine-readable prerequisite DAG as the sole curriculum specification.

3.4 LLM-Generated Educational Content

The closest recent prior art is the emerging category of LLM-based curriculum generators. Microsoft's Phi-1 paper (Gunasekar et al., 2023) demonstrated that GPT-3.5-generated "textbook quality" synthetic data dramatically outperforms raw web data for training small LLMs — establishing that LLMs can produce pedagogically useful text, but using it as training data rather than as a human-readable product, and with no prerequisite structure constraining content order. Chen et al. (IEEE TLT, 2024) use GPT-4 with self-critique loops to generate lesson plan components; a 2025 multi-LLM agent system (arXiv:2507.05528) adds a reviewer agent as a soft pedagogical quality gate. These are the most structurally similar to our GENERATE $\rightarrow$ EVALUATE $\rightarrow$ refine loop, but neither uses a prerequisite DAG to determine what to generate or in what order.

One 2025 paper (arXiv:2501.12300) uses LLMs to *complete* a curriculum knowledge graph for course recommendation — the closest work to combining KGs with LLMs for education — but the graph drives recommendation, not generation of textbook prose.

3.5 Machine-Readable Curriculum Standards

IMS Learning Design (2003) is the most direct institutional precedent for a machine-readable curriculum specification: an XML standard encoding roles, resources, and activity flows in a "Unit of Learning" that an LD-aware player can execute. IMS LD demonstrates the value of separating the *specification* from the *execution engine* — the same DODA insight that drives SPL's design. Its limitation is the pre-LLM assumption: IMS LD orchestrates pre-authored resources; it cannot generate content. The proposed system inherits IMS LD's architectural insight and adds LLM generation as the execution layer.

A 2025 OWL/RDF curriculum ontology (arXiv:2506.05751) maps prerequisite DAGs to triple-stores for recommendation, and K12-KGraph (arXiv:2605.09635) extracts a curriculum-aligned concept graph from Chinese K-12 textbooks for LLM benchmarking. Both confirm the value of machine-readable prerequisite structure but use it for retrieval or evaluation, not generation.

3.6 Morpheme-Based Language Learning

The Chinese Characters and English Morphology domains rest on a decades-old pedagogical claim: vocabulary acquisition is faster when learners know the component morphemes. Zhang (2016) provides psycholinguistic evidence that radical decomposition knowledge directly predicts L2 Chinese vocabulary size. Li et al. (EMNLP 2015) show computationally that radical-aware embeddings improve NLP tasks, confirming that radical decomposition encodes genuine semantic structure. WikiMorph (NSF, 2021) provides a GPT-2-based morpheme decomposer for English. None of these systems pair the morpheme graph with LLM generation of structured lessons — that combination is the contribution of the Chinese Characters and English Morphology domains in the proposed system.

3.7 Summary

The literature reveals a clear gap. Prerequisite-graph research exists but produces mined, noisy graphs and no content. LLM content-generation research exists but lacks graph-structured ordering and formal verification. ITS systems achieve high learning gains but require per-domain expert authoring at prohibitive cost. KST and ALEKS address assessment, not generation. No prior system combines: (1) a human-authored, machine-readable prerequisite DAG as the sole curriculum specification; (2) an LLM workflow that consumes that DAG to generate a complete, prerequisite-ordered textbook; (3) a formal verifier as a quality gate on generated content; and (4) a single unified framework demonstrated across genuinely heterogeneous domains. The proposed system occupies this gap.

4 The Concept-Graph Schema

4.1 Schema Structure

A domain graph has four elements. The snippet below uses the current reference serialization (YAML) purely for illustration; this is the *compiled representation* produced by the underlying tools, not the user-facing authoring interface. The concept-graph itself is a topological abstraction — the SPL workflow does not care whether it was drawn in a GUI, expressed in a structured file, or generated from a domain expert interview. Only the graph's structure matters at runtime:

```
domain: classical_mechanics

primitives:
  position:
    symbol: x
    defines: "first radical — configuration; where things are"
    tier: 0
  time:
    symbol: t
    defines: "first radical — the parameter motion unfolds in"
    tier: 0
  mass:
    symbol: m
    defines: "second radical — inertia; resistance to changes in motion"
    tier: 0
  force:
    symbol: F
    defines: "second radical — interaction; what changes states of motion"
    tier: 0

first_radical_primitives: [position, time]

concepts:
```

```
  velocity:
    composed_of: [position, time]
    defines: "rate of change of position — v = dx/dt"
    verifier: sympy
    lab: sympy.diff
    play: "differentiate x(t) = t**2; compare v(t) against the difference quotient"
    tier: 1
  momentum:
    composed_of: [mass, velocity]
    defines: "p = m • v — the conserved currency of motion"
    verifier: sage|sympy
    lab: "graph_lib.verify_momentum_conservation"
    tier: 2
  ...
  normal_modes:
    composed_of: [eigenvalue, superposition, constraints]
    defines: "the natural oscillation frequencies of a coupled system"
    verifier: sympy
    tier: 5

applications:
  orbital_mechanics: {uses: [normal_modes, conservation_of_energy]}
  robotics:          {uses: [constraints, lagrangian_mechanics]}
```

The schema has four elements:

Primitives are the irreducible brick set — concepts with no prerequisites. Each carries a `symbol`, a one-sentence `defines`, and a `tier` (always 0). The schema distinguishes `first_radical_primitives` (the FORM class — concepts reducible from these alone) from the remaining primitives (the SOUND/operator class — genuinely independent content). This split encodes a structural theorem of the domain.

Concepts form a directed acyclic graph via `composed_of` edges. Each concept carries a `verifier` tag that specifies which verification backend to invoke (`sympy`, `sage`, `sage|sympy`, `lean`, `lean|sympy`, `structural`, or a named Python tool). The optional `lab` and `play` fields give an executable worked example and a student exercise.

`first_radical_primitives` is the curated subset encoding the irreducibility theorem: `graph_lib.reducible(first_rad` must return `False` for a well-formed domain. This is the machine-checked version of "you cannot teach this domain from one class of primitives alone."

Applications list the downstream use-cases that the capstone concept unlocks. These populate the capstone "Payoff" section of the generated book.

4.2 The Two-Radical Structure

Every domain in our corpus exhibits a two-radical structure: two mutually irreducible classes of primitives, each necessary, neither sufficient alone.

Domain	First radical (FORM)	Second radical	Irreducibility theorem
Classical Mechanics	position, time (kinematics)	mass, force (dynamics)	No kinematic quantity predicts F=ma
Chinese Characters	11 semantic pictograms	马 (phonetic lender)	Pictograms alone cannot yield phono-semantic compounds
English Morphology	5 roots (spect, port, …)	prefixes + suffixes	Roots alone cannot yield *inspection*
Chemistry Elements	8 elements (H,C,N,…)	valence_electron	Element identity alone does not predict bonding

Domain	First radical (FORM)	Second radical	Irreducibility theorem
Music Theory	pitch, semitone (pitch space)	beat, halving (rhythm)	Pitch alone cannot yield rhythm structures

This is not a design choice imposed by the schema — it is a structural fact discovered in the domain and made checkable by `graph_lib.reducible()`. The schema makes the fact explicit; the algorithm verifies it.

4.3 The LEGO Payoff Shape

Table 1. Ten domains: primitive count, concept count, capstone, verifier family, and the LEGO payoff.

Domain	Primitives	Concepts	Capstone	Verifier	LEGO Payoff

Symbolic / Algorithmic — exact computation, formal verification

Domain	Primitives	Concepts	Capstone	Verifier	LEGO Payoff
Linear Algebra	5	25	Spectral Theorem	SymPy	5 primitives → ML, quantum, signal processing
Introductory Geometry	5	19	Trigonometric Ratios	SymPy / Sage	5 primitives → navigation, architecture, graphics
Classical Mechanics	4	19	Normal Modes	SymPy / Sage	4 primitives → robotics, orbital mechanics, quantum bridge
SageMath	4	18	Experimental Mathematics	Sage / SymPy	4 primitives → CAS-aided mathematics
Lean Theorem Proving	4	19	AI-Assisted Proving	Lean / SymPy	4 primitives → formal proof, AI mathematics
SQL & Relational Algebra	5	~16	Execution Plan Tuning	Structural / SymPy	5 primitives → any query optimization problem *(planned)*
Quantum Mechanics	5	~18	Deutsch-Jozsa Algorithm	SymPy Quantum	5 primitives → quantum computing, QFT *(planned)*

Network-Driven / Balanced — hierarchical composition, conservation laws

Domain	Primitives	Concepts	Capstone	Verifier	LEGO Payoff
Chemistry Elements	9	14	Stoichiometry	Structural	9 bricks → every formula on every label
Python for Science	4	19	Reproducible Science	NumPy/SymPy/SciPy-driven	4 primitives → full scientific Python stack
Central Dogma (Molecular Biology)	5	~16	Protein Synthesis Regulation	Structural	5 primitives → gene expression, biotechnology *(planned)*

Morphological / Structural — decomposition, composition rules, pattern recognition

Domain	Primitives	Concepts	Capstone	Verifier	LEGO Payoff
Chinese Characters	12	16	Phono-semantic Principle	Structural	~200 radicals → thousands of characters
English Morphology	13	18	Morphological Decoding	Structural	~220 morphemes → most academic vocabulary
Music Theory	4	17	Four-Chord Loop	Structural	4 primitives → most popular music

The 10 implemented domains validate the framework; the 3 planned domains (SQL, Quantum Mechanics, Molecular Biology) are pure data-authoring tasks — new concept-graph files, zero new code — demonstrating that the schema scales horizontally without bound.

The payoff column is not aspirational copy — it is the `applications` section of the domain graph, which the SPL workflow uses to generate the capstone "Payoff" section of the book.

5 From Graph to Book: The SPL Workflow

5.1 `build_concept_book.spl` — One Source, Any Domain

The entire content-generation pipeline is encoded in a single SPL workflow:

```
WORKFLOW build_concept_book
    INPUT:
        @domain_yaml TEXT DEFAULT 'linalg_graph.yaml',
        @target      TEXT DEFAULT 'spectral_theorem',
        @style       TEXT DEFAULT 'textbook',
        @language    TEXT DEFAULT 'en',
        @output_dir  TEXT DEFAULT ""
    OUTPUT:
        @textbook TEXT
DO
    -- 1. Load graph, validate (acyclicity + reducibility), compute teaching order
    CALL setup_domain(@domain_yaml, @target, @payoff_weight) INTO @_
    CALL order_length(@domain_yaml) INTO @order_len
```

```
    -- 2. Generate and verify one section per concept in dependency order
    @_i := 0
    WHILE @_i < @order_len DO
        CALL order_item(@domain_yaml, @_i) INTO @concept

        CALL cache_get(@concept) INTO @section
        EVALUATE @section
            WHEN = "miss" THEN
                GENERATE write_section(@concept, ...) INTO @section

                -- Primitive budget: at most one new primitive per section
                CALL needs_primitive_refinement(...) INTO @_refine
                EVALUATE @_refine
                    WHEN = "yes" THEN
                        GENERATE refine_section(@section, "one primitive per section") INTO @section
                END

                -- Domain verifier: structural / sympy / sage / lean
                CALL verify_section(@section, @domain_yaml) INTO @check
                EVALUATE @check
                    WHEN contains("fail") THEN
                        GENERATE refine_section(@section, @check, @style_guide) INTO @section
                END
                CALL cache_put(@concept, @section) INTO @_
            ELSE
                LOGGING "cache HIT — 0 LLM calls" LEVEL INFO
        END
        @textbook := @textbook + @section
        @_i := @_i + 1
    END

    -- 3. Capstone payoff section
    CALL apps_list(@domain_yaml) INTO @apps
    GENERATE write_payoff(@target, @apps, @style_guide) INTO @capstone
    @textbook := @textbook + @capstone

    COMMIT @textbook
END
```

The only domain-selecting input is `@domain_yaml`. Everything else — the graph load, the teaching order, the verifier dispatch, the book assembly — is driven by the concept-graph content. Switching from Linear Algebra to Classical Mechanics to Chinese Characters requires changing one parameter.

5.2 Teaching Order as a Graph Algorithm

`setup_domain()` performs three steps before any LLM call:

1. **Load**: `graph_lib.load_domain(yaml_path)` parses the YAML into a `networkx.DiGraph` where edges point from prerequisites to dependents.
2. **Validate**: `graph_lib.acyclic(graph)` asserts no cycles; `graph_lib.reducible(graph, primitives)` asserts every concept traces back to the declared primitive set. If either check fails, the workflow raises `ToolFailed` before any generation begins.
3. **Order**: `graph_lib.productivity_order(graph, target, payoff_weight)` produces the teaching sequence — a topological ordering biased toward concepts that are ancestors of the target, weighted

by how many concepts they unlock.

The priority score $P(v)$ for each concept node v is:

$$P(v) = \alpha \cdot |\text{descendants}(v, \text{target})| - d(v, \text{target})$$

where $\text{descendants}(v, \text{target})$ is the number of nodes on paths from v toward the capstone target, $d(v, \text{target})$ is the shortest-path distance from v to the target in the DAG, and α is the `payoff_weight` parameter. High α biases the order depth-first toward the capstone; low α biases breadth-first to maximize primitive coverage early. Nodes are emitted in descending $P(v)$ order, subject to topological constraints (no concept is emitted before its prerequisites).

The teaching order is deterministic and reproducible. Given the same concept-graph and the same α, two runs on different machines produce identical sequences.

5.3 Verifier Dispatch

Each concept's `verifier` tag determines what happens after generation:

Tag	Backend	What it checks
`sympy`	SymPy CAS	Algebraic identity, derivatives, integrals — recomputed over symbolic expressions
`sage|sympy`	SageMath (preferred) / SymPy (fallback)	Exact arithmetic — momentum conservation, energy ledgers, equation balancing
`lean|sympy`	Lean 4 + mathlib	Formal proof — type-checks the section's logical claims
`structural`	`graph_lib.verify_character_lego`	Graph-as-oracle — checks that every composite listed in the section decomposes correctly into declared primitives
`numpy`, `scipy`, `pandas`, `sklearn`, `matplotlib`	Named Python tool	Code-execution check against the scientific Python stack

The SPL call is identical across all verifier types:

```
CALL verify_section(@section, @domain_yaml) INTO @check
```

`graph_lib.verify_content()` reads the concept's `verifier` field from the `domain_data` and dispatches to the appropriate backend. A new domain plugs in a new verifier branch inside `verify_content` — never a new SPL construct.

5.4 Content Cache

The content cache (`graph_lib.cache_get` / `cache_put`) stores verified sections keyed by concept name and domain. On a cache hit, the section is reused with zero LLM calls. This matters in two scenarios: regenerating the book with a different `@style` or `@language` (misses all sections — full generation run), and iterating on a partially-generated book where a later concept's verifier failed (hits all earlier concepts — generation resumes from the failure point). Across ten domains, the cache reduces total generation cost substantially on any run after the first.

5.5 DODA for Education

SPL's DODA principle (Declarative Once, Deploy Anywhere) means the same `build_concept_book.spl` runs unchanged on any model backend:

```
# Cloud: Anthropic claude-sonnet-4-6
spl3 run build_concept_book.spl \
  --adapter claude_cli -m claude-sonnet-4-6 \
  --param domain_yaml=mechanics_graph.yaml --param target=normal_modes

# Local: Ollama + gemma4
spl3 run build_concept_book.spl \
  --adapter ollama -m gemma4 \
  --param domain_yaml=chinese_characters_graph.yaml --param target=phono_semantic_principle

# Multilingual: same workflow, French output
spl3 run build_concept_book.spl \
  --param domain_yaml=linalg_graph.yaml --param language=fr
```

The curriculum (concept-graph) and the workflow (SPL) are orthogonal to the model choice. A teacher who authors a new domain graph can deploy it against any available model without touching the workflow.

6 The Concept-Book Portal

6.1 Architecture

The system has two layers:

Content pipeline (SPL.py): `build_concept_book.spl` runs against a domain the concept-graph and produces two artifacts: a `{domain}_graph.html` standalone vis.js concept navigator, and a `{domain}_concept_book.html` self-contained HTML book with MathJax rendering and a two-column TOC-sidebar layout.

Web portal (concept-book): A Vite + Vanilla JS frontend serves a domain grid on the home page. Clicking a domain loads a split view: the vis.js graph navigator (left, in an iframe) and a concept detail panel (right). A "Generate Book" sidebar lets the user select a target concept and stream `spl3` output live via Server-Sent Events through a FastAPI backend.

Generated books are written to `public/domains/{id}/concept_book.html` and served as static files. The portal deploys to GitHub Pages with `npm run deploy`; the FastAPI backend is a local tool not included in the static deployment.

6.2 Interactive Graph Navigation

The vis.js navigator renders the concept DAG with primitives at the top (tier 0) and the capstone at the bottom (highest tier). Clicking any concept node highlights its `learning_path` — the minimal prerequisite sequence from the primitive set to that concept — and displays its `defines`, `lab`, and `play` fields in a sidebar panel.

This turns the graph into a learner-facing navigation instrument: the student can see not just *what* a concept is, but *why it appears where it does* in the learning sequence, and *what they need to know first*.

6.3 Ten Domains, One Portal

[Figure 1: concept-book portal — domain grid home page and a graph navigator view for Classical Mechanics, with the learning path to "normal_modes" highlighted.]

The breadth of the ten domains in a single portal makes the central claim visceral: the same UI, the same workflow, the same graph schema — applied to linear algebra and Chinese characters and music theory simultaneously. The domain grid groups domains by tag (math, physics, computing, language, arts), but the underlying structure is identical across all of them.

7 Evaluation

7.1 Schema Generalization: Regression Oracle

The generalization from domain-specific Python modules (recipes 71, 73) to the domain-agnostic concept-graph schema is validated by `validate_graph_lib.py`, which runs 40 comparative checks between the graph-driven path (`graph_lib.load_domain` + `graph_lib.build` + `graph_lib.*`) and the frozen per-domain Python modules for Linear Algebra and Geometry. All 40 checks pass:

```
ALL CHECKS PASSED — graph_lib.py + {domain}_graph.yaml  frozen domain modules
```

This is the machine-checked proof that the generalization is behaviorally identical, not merely similar.

7.2 The Reducibility Theorem Across Domains

For each domain, `graph_lib.reducible(graph, first_radical_only)` returns `False` — confirming that the second radical class is genuinely irreducible content. This is the formal counterpart to the pedagogical claim that no subset of the primitives is sufficient on its own.

Domain	`reducible(all_primitives)`	`reducible(first_radical_only)`
Classical Mechanics	True	False — dynamics cannot be derived from kinematics
Chinese Characters	True	False — 马 (phonetic) is not derivable from 11 pictograms
English Morphology	True	False — affixes are not derivable from roots
Chemistry Elements	True	False — valence_electron is not derivable from element identities
Music Theory	True	False — rhythm is not derivable from pitch

The reducibility check runs in milliseconds on graph load — before any LLM call — and raises `ToolFailed` if the declared graph does not satisfy it. This prevents a malformed domain graph from silently generating incorrect content.

7.3 Teaching-Order Stability

`productivity_order` with varying `payoff_weight` (0.5, 1.0, 1.5, 2.0) produces stable but adjustable teaching sequences. At low weights, the order is closer to a breadth-first traversal (maximizing primitive exposure early); at high weights, it biases toward the capstone's direct ancestor chain. Across all ten domains, the relative order of concepts within a tier is stable across weights; only cross-tier interleaving shifts.

7.4 Content Verification Pass Rates

[Table: per-domain verifier pass rate on first generation vs. after one refinement pass — to be populated from SPL.py run logs.]

The primitive budget constraint (at most one new primitive introduced per section) is the dominant source of first-generation refinement requests in mathematical domains. In structural domains (Chinese characters, English morphology), the verifier pass rate on first generation is higher because the structural check is binary and the LLM has been prompted to follow the decomposition schema.

8 Discussion

8.1 Curriculum Knowledge is Graph-Structured

The ten domains in this paper were not chosen to validate a prior belief — they were the domains for which concept-graphs were authored, in the order they were authored. The fact that all ten exhibit the same LEGO payoff shape suggests this shape is a property of *how knowledge is organized*, not of our selection. A domain without a finite primitive set and a composition grammar is harder to teach systematically — and harder to learn.

Making the graph explicit does not change what the domain is. It changes what becomes *computable* about it: the teaching order, the prerequisite path to any concept, the reducibility of the whole from the parts, and the machine- checkable verification of each section. These were always present in expert teaching; the concept-graph makes them available to anyone with domain knowledge — regardless of how they choose to express that graph.

8.2 Language Learning as a STEM Discipline

The Chinese characters and English morphology domains share a verifier (`verify_character_lego`) that checks decomposition correctness. The same function handles 林 = 木 + 木 and *inspection* = in- + spect + -ion, because both are instances of the same abstract operation: verify that the declared `composed_of` pieces are all present in the declared primitive set. The schema does not know it is handling two writing systems — it handles one abstraction.

This is a non-trivial claim: vocabulary acquisition in any morphologically compositional language is, formally, the same problem as chemical formula reading or character decomposition. The LEGO payoff shape makes the analogy precise: learn the bricks, learn the grammar, decode the space.

8.3 Limitations

Graph authoring requires domain expertise. The schema is easy to write; knowing *what the primitives are* and *what counts as composition* requires substantive domain knowledge. For some domains (music theory, Lean proving), the primitive selection involves genuine pedagogical judgment.

Verifiers have a cold-start cost for complex domains. Deploying Lean 4 or SageMath as verifiers requires non-trivial environment setup that may be a barrier for educators without systems experience. The key mitigation is that `verifier: structural` serves as a universal zero-setup baseline for any new domain: it checks only that every concept's `composed_of` pieces are present in the declared primitive set — no CAS or proof assistant required. A domain author can bootstrap with structural verification and upgrade to symbolic verifiers incrementally. A future direction is LLM-generated verifier stubs, where the system auto-generates a `verify_content` branch from the domain YAML before a hand-crafted verifier is available.

LLM-generated prose is fluent but not unconditionally correct. The verifier checks structural or symbolic claims; it does not evaluate every sentence. A domain without a symbolic verifier can use `verifier: structural` as a minimum quality gate, accepting that semantic accuracy depends on the model's fluency rather than machine-checked computation.

The portal is currently local-first. Book generation requires a running `spl3` backend with a model adapter configured. The static portal (GitHub Pages) serves pre-generated books but does not support on-demand generation without a local backend.

8.4 Future Work

`answer_on_demand.spl`: personalized lessons using `graph_lib.gap(known, target)` to identify which concepts a learner still needs, generating only the missing sections. This turns the system from a book-generator into a personal tutor.

Multi-language books: the `@language` parameter is already wired into `build_concept_book.spl` (ISO 639-1 codes; mathematical notation is language- invariant). Generating the same book in English, Chinese, and French requires one parameter change.

Community graph authoring: a shared registry of domain concept-graphs, where any domain expert can author a graph — via file, GUI, or LLM interview — and publish a concept-book without writing code.

Concept-Net: a cross-domain network connecting concepts across graphs (e.g., *eigenvalue* in the Linear Algebra graph appears in the Classical Mechanics graph as a dependency of *normal_modes*). Cross-domain edges would enable learning paths that span subjects.

Assessment generation: the `lab` and `play` fields in each concept node are already structured as executable exercises. An assessment workflow would generate graded problem sets from the graph, ordered by tier.

9 Conclusion

Teaching any domain is graph authoring. A concept-graph — specifying primitive concepts, composition relationships, per-concept verifiers, and a capstone — is a complete curriculum specification, independent of how it is expressed or authored. A single SPL workflow converts any such graph into a verified, prerequisite-ordered, interactively navigable textbook, with no code changes between domains. We demonstrated this across ten domains spanning mathematics, physics, computing, natural languages, and music. The central structural property that makes this possible — the LEGO payoff shape, machine-checked by the reducibility theorem — is present in all ten and is, we argue, a general feature of well-organized knowledge.

The contribution is not a better LLM prompt or a more capable model. It is a *representation*: the concept-graph that makes curriculum structure explicit, computable, and verifiable. Once the graph exists, teaching becomes a graph algorithm and learning becomes pathfinding.

10 References

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Appendix A: Working Plan

This section tracks open revision items. Remove before final submission.

Completed

- ☒ Paper structure: 8 main sections + Appendix A (Working Plan)
- ☒ Title confirmed: "ConceptBook: A Graph-First Framework for AI-Generated Curricula"
- ☒ Abstract rewritten: introduces ConceptBook by name; notes authoring-tool independence
- ☒ §1.3 rewritten: physicist's reductionism → ZiNets → universal curriculum arc; mission statement added
- ☒ YAML de-emphasized throughout: concept-graph is the contribution; YAML is one serialization
- ☒ Related Work (§2): 7 subsections, 25+ papers, cross-cutting comparison table, gap statement
- ☒ Build pipeline: `build_md2tex.sh`, `build_tex2pdf.sh`, `polish_tex.py` — xelatex (CJK support)
- ☒ Venue updated: AIED 2027 primary, TMLR secondary (pdflatex caveat noted); NeurIPS 2026 passed
- ☒ First PDF compiled: 14 pages
- ☒ **v0.2 — Gemini feedback incorporated:**
  - ☒ Removed all hard-coded numeric labels from headings (LaTeX auto-numbers)
  - ☒ §1.1: "vibe tutoring" contrast added
  - ☒ §3.1: YAML-as-compiled-serialization clarification added
  - ☒ §4.2: formal $P(v)$ equation for `productivity_order` added
  - ☒ §7.3 Limitations: cold-start verifier section expanded; `structural` as universal baseline; LLM verifier stubs mentioned
  - ☒ Table 1 expanded to 13 domains (SQL, Quantum Mechanics, Molecular Biology added as planned)
  - ☒ Table 1 grouped by structural behavior: Symbolic/Algorithmic, Network-Driven/Balanced, Morphological
  - ☒ Venue table updated: MLSys 2027 (Oct 2026) added

In Progress

- ☐ **Abstract** — tighten to 200 words; "vibe tutoring" framing not yet in abstract

Pending — Content

- ☐ **Verifier pass-rate table (§6.4)** — run `spl3` against all 10 domains and collect: first-generation pass rate, refinement rate, cache hit rate per domain; populate Table 4
- ☐ **Figure 1** — concept-book portal screenshot (domain grid + graph navigator); export from running portal at `http://localhost:5174/concept-book/`
- ☐ **Figure 2** — concept DAG visualization for Classical Mechanics or Chinese Characters; export from vis.js (`graph.html`) as PNG/SVG
- ☐ **Figure 3** — SPL workflow diagram (concept-graph → teaching order → GENERATE → verify → cache → book); draw as architecture diagram
- ☐ **Figure 4** — LEGO payoff chart (primitive count vs. generative space, across 10 domains); matplotlib bar/scatter
- ☐ **Teaching-order stability table (§6.3)** — run `productivity_order` at payoff_weight 0.5/1.0/1.5/2.0 for 2–3 domains; show sequence stability
- ☐ **§5 Portal section** — add actual screenshot description or figure reference once Figure 1 exists
- ☐ **User study (optional, strengthens AIED/TMLR)** — compare ConceptBook-generated book vs. human-authored curriculum on a domain; even informal feedback from physicist friend counts as qualitative evidence

Pending — Writing

- ☐ **Proof-read full paper** — check flow, section transitions, and consistency
- ☐ **Fill [CITE-*] placeholders** — verify full author lists, venues, years for all references in §2

- ☐ **Tighten §4 (SPL Workflow)** — the SPL code excerpt is long; consider moving part to an appendix
- ☐ **Add Appendix B: Domain Graph Schema Reference** — full YAML schema with field descriptions; allows §3 to be shorter
- ☐ **Add Appendix C: Graph Algorithm Reference** — `productivity_order`, `learning_path`, `gap`, `reducible` with pseudocode

Pending — Venue

- ☐ **AIED 2027 style files** — monitor aied2027.org; swap venue style block in `build_md2tex.sh` when `llncs.cls` becomes available (~late 2026); deadline est. Feb 2027
- ☐ **MLSys 2027** — deadline est. Oct 2026; reframe emphasis on SPL engine infrastructure, primitive budget, and deterministic execution if targeting this venue
- ☐ **Page count check** — AIED 2027 main track is typically 12 pages + references; expect ~10–11 pages in LNCS format; trim SPL workflow code excerpt if needed
- ☐ **TMLR option** — if targeting TMLR: (a) romanize inline Chinese characters with first-use gloss, (b) switch to TMLR pdflatex pipeline, (c) add user study or comparative evaluation
- ☐ **Planned domains** — author `sql_graph.yaml`, `quantum_graph.yaml`, `molecular_biology_graph.yaml` to bring total to 13; each is pure data authoring, zero new code
- ☐ **Remove this appendix** before final submission